A POSITIVE WEIGHTING IS UPPER SEMI-CONTINUOUS

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 22678-22692, immediately following the statement at 22671-22676.

Editorial note. The source line 22686 contains the author's duplicated word 'we we'; it is retained verbatim.

Editorial note. One chapter lemma and the definition of a weighting are cited by \ref as navigation pointers.

Allowed standard prerequisites (editorial).

- Etale neighbourhood making the quasi-finite part finite with a unique point. chapter lemma lemma-etale-makes-quasi-finite-finite-multiple-points-var, cited at proof line 22683. Produces V -> Y etale, U open in X_V, finite U -> V and the unique point u over v with purely inseparable residue extension.
- Definition of a weighting and of its integral. chapter definition definition-weighting, cited at proof line 22685. Gives the local constancy of the integral and the fact that the weighting of any point of U appears as a summand of the integral at the image point.
- Openness of the etale morphism U -> X. standard Stacks fact used at proof line 22690. Transfers the upper bound established on U back to a neighbourhood of x in X.

Source reference. Original definition 0F3A: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 22152–22152. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 02LM: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 12019–12019. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $f : X \to Y$ be a locally quasi-finite morphism. Let $w : X \to \mathbf{Z}_{>0}$ be a positive weighting of f. Then w is upper semi-continuous.*

Proof. Let $x \in X$ with image $y \in Y$. Choose an étale neighbourhood $(V, v) \to (Y, y)$ and an open $U \subset X_V$ such that $\pi : U \to V$ is finite and there is a unique point $u \in U$ mapping to v with $\kappa(u)/\kappa(v)$ purely inseparable. See Lemma 02LM. Then $(\int_\pi w|_U)(v) = w(u)$. It follows from Definition 0F3A that after replacing V by a neighbourhood of v we we have $w|_U(u') \leq w|_U(u) = w(x)$ for all $u' \in U$. Namely, $w|_U(u')$ occurs as a summand in the expression for $(\int_\pi w|_U)(\pi(u'))$. This proves the lemma because the étale morphism $U \to X$ is open. □

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.